**PATEROS TECHNOLOGICAL COLLEGE**
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Institute of Information and Computing Technology

# i-AdminConnect: A Web-Based Administrative Services and Records Tracking System for Teachers and Students

A Capstone Project
Submitted to the Faculty of the
Institute of Information and Computing Technology

In Partial Fulfillment
of the Requirements for the Degree of
Bachelor of Science in Information Technology

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# CHAPTER I

# INTRODUCTION

## 1. Background of the Study

### i-AdminConnect: A Web-Based Administrative Services and Records Tracking System for Teachers and Students

Educational institutions continue to rely on administrative services to support students, teachers, and school personnel in carrying out academic and institutional processes. These services include processing document requests, managing records, monitoring transactions, and providing timely communication between stakeholders. In many schools, these processes are still handled manually or through disconnected systems, resulting in delays, misplaced records, duplicated data, and difficulties in tracking the status of requests.

The increasing adoption of information technology has transformed how educational institutions manage administrative operations. Web-based information systems provide an efficient platform for automating routine tasks, improving data accessibility, and enhancing communication among users. By centralizing records and services into a single online system, schools can reduce paperwork, minimize human error, and improve the quality of services provided to teachers and students.

The proposed **i-AdminConnect: A Web-Based Administrative Services and Records Tracking System for Teachers and Students** aims to provide an integrated platform where students can submit service requests, monitor the progress of their requests, and access authorized records. Administrators can efficiently process requests, maintain records, generate reports, and monitor overall system activities through a secure role-based environment.

The system will also incorporate real-time request tracking features that allow users to monitor the status of their transactions. Automated notifications, centralized records management, dashboard analytics, and reporting tools will improve transparency, accountability, operational efficiency, and administrative decision-making. These features also aim to reduce paperwork, minimize lost or misplaced requests, and lessen the need for teachers and students to repeatedly visit the administrative office for follow-up. Through the implementation of this system, the institution can provide faster, more accurate, and more reliable administrative services while supporting digital transformation within the school.

**2. Rationale of the Study**

Educational institutions continuously manage a wide range of administrative services that support the daily operations of the school and the needs of students, teachers, and administrative personnel. These services include processing document requests, maintaining institutional records, monitoring transactions, and providing timely updates to stakeholders. In many schools, these processes are still performed manually or through separate communication channels, resulting in delayed processing, difficulty in tracking requests, duplication of records, and increased administrative workload.

The rapid advancement of information technology has encouraged educational institutions to adopt web-based systems that streamline administrative operations and improve service delivery. A centralized web-based platform allows users to submit requests, monitor transaction status, and access authorized information anytime and anywhere with an internet connection. At the same time, administrative personnel can efficiently organize records, process requests, and generate reports while minimizing manual work and reducing the possibility of human error.

The development of i-AdminConnect: A Web-Based Administrative Services and Records Tracking System for Teachers and Students is proposed to address these challenges by providing an integrated platform for managing administrative services and institutional records. The system is designed to improve efficiency, transparency, and accessibility by allowing students and teachers to submit and monitor requests online while enabling administrators to manage records and transactions through a secure role-based system.

Furthermore, the proposed system supports the digital transformation initiatives of educational institutions by promoting organized record management, faster service delivery, improved communication, and accurate reporting. Through the implementation of i-AdminConnect, the

institution can enhance the quality of administrative services, reduce processing time, and provide a more reliable and convenient experience for all authorized users.

**Objectives of the Study**

**1. General Objective**

To design, develop, and implement i-AdminConnect: A Web-Based Administrative Services and Records Tracking System that enhances the efficiency of administrative services and improves the management and tracking of institutional records for teachers and students.

**2. Specific Objectives**

Specifically, the study aims to:

1. Develop a secure web-based administrative services and records tracking system for teachers, students, and administrators.
2. Implement a secure user authentication and access management mechanism to ensure that administrators, teachers, and students can securely access only the features and records appropriate to their roles.
3. Develop a centralized records management module for storing, updating, retrieving, and organizing administrative documents and requests.
4. Implement a real-time request tracking feature that allows teachers and students to monitor the status of their requests, reducing unnecessary follow-up visits and improving communication with the administrative office.
5. Provide automated notifications to inform teachers and students whenever there are updates on the status of their requests, helping minimize delays and unnecessary trips to the school.
6. Generate automated reports, transaction histories, and dashboard analytics that support efficient monitoring, accountability, and administrative decision-making while maintaining accurate records of all processed and released requests.
7. Reduce paperwork, minimize the risk of lost or misplaced requests, and improve the efficiency, transparency, and accessibility of administrative services through a centralized web-based platform.

**1. Scope of the Study**

This study focuses on the design, development, and implementation of i-AdminConnect: A Web-Based Administrative Services and Records Tracking System for Teachers and Students. The proposed system is intended to improve the delivery of administrative services and the management of institutional records within a single educational institution by providing a centralized and secure web-based platform.

The system is designed for three types of users: administrators, teachers, and students. It provides secure user authentication and role-based access control to ensure that each user can access only the functions and information appropriate to their assigned role.

For students, the system enables the online submission of administrative service requests, including requests for Good Moral Certificates, Certificates of Grades, Certificates of Enrollment, and other authorized academic or administrative documents. Students can monitor the status of their requests in real time, receive automated system notifications regarding request updates and document availability, track the processing history of their requests, and view the current status of their submitted requests through the system. Once approved, students will be notified when their requested documents are ready for claiming at the administrative office.

For teachers, the system facilitates the submission and tracking of administrative requests, including requests for Good Moral Certificates, Service Records, Certificates of Ratings, Certificates of Employment, leave-related documents, and other authorized records, as well as the receipt of transaction updates and notifications. For administrators, the system provides tools for processing requests, managing institutional records, monitoring transactions, generating reports, maintaining user accounts, and recording user activities through audit logs.

The system also includes records management and records tracking modules that allow authorized users to store, update, retrieve, and monitor the progress of administrative requests from submission to completion. A dashboard provides an overview of ongoing transactions, pending requests, completed requests, and other relevant administrative information to support efficient monitoring and decision-making.

The study is limited to administrative services and records management functions within the selected educational institution. It does not include online payment processing, learning management functionalities, student grading, financial accounting, payroll management, or other academic information system modules. The system requires an internet connection and can only be accessed by authorized users with valid login credentials.

A dashboard with analytics provides real-time statistics on pending, approved, completed, and rejected requests. It also presents graphical reports, monthly transaction summaries, document request trends, and user activity to assist administrators in monitoring administrative operations and making informed decisions.

**Limitations of the Study**

This study has certain limitations that may affect its overall scope and implementation.
• *What features are not included?*

1. **Features Not Included**
The system does not cover the following features:

· It does not include online payment processing for administrative services or document requests.

· It does not provide Learning Management System (LMS) functionalities such as online classes, assignment submission, quizzes, or examinations.
· It does not include student grading, class scheduling, enrollment management, or financial accounting modules.
· It does not support electronic document signing or digital signature verification.
· It does not provide mobile application support, as the system is designed to operate through a web browser.
· It does not integrate with external government or educational databases, and all records are managed only within the authorized educational institution.

• ***What factors may affect the system?***

The performance and effectiveness of the system may be affected by internet connectivity, since users must have a stable internet connection to access the web application and perform administrative transactions.

The accuracy and reliability of records also depend on the correctness and completeness of the information entered by administrators, teachers, and students. Incorrect or incomplete data may result in inaccurate records or delays in processing requests.

The successful implementation of the system also depends on the cooperation of school administrators, teachers, and students in properly using the system and following the established administrative procedures.

• ***What are the technical constraints?***

The system is limited by the following technical constraints:

- ·Dependence on a web-based server, hosting environment, and database management system.
- · Browser compatibility and device performance may affect the user experience.
- ·Storage capacity and system performance depend on the available server resources and database configuration.
- ·The system requires regular maintenance, software updates, and database backups to ensure continuous operation and data security.

• ***What conditions are beyond the control of the researchers?***

There are also conditions that are outside the control of the researchers, including:

- · Power interruptions and internet service outages that may temporarily prevent users from accessing the system.
- · Hardware failures such as server downtime, computer malfunctions, or network equipment failures.
- ·Changes in institutional policies, administrative procedures, or document requirements after the implementation of the system.
- ·Unauthorized access attempts, cybersecurity threats, or other external security risks despite the implementation of appropriate security measures.
- ·Human errors, including incorrect data entry, accidental deletion of records, or improper use of system functions by authorized users.

**Conceptual Framework of the Study**

The study is anchored on the Input–Process–Output (IPO) Model, which serves as the conceptual framework for the development of the proposed i-AdminConnect: A Web-Based Administrative Services and Records Tracking System for Teachers and Students. The IPO model illustrates the relationship among the resources required for the study (Input), the activities involved in designing and developing the system (Process), and the expected outcomes after the system's implementation (Output).

The Input consists of the necessary resources and requirements needed to develop the system. These include user information such as student, teacher, and administrator accounts; administrative service requests; institutional records; school policies and procedures; and the required hardware and software resources. Functional requirements, including secure user authentication, role-based access control, request management, records management, notification services, dashboard analytics, and report generation, also form part of the input stage.

The Process represents the activities undertaken in developing the proposed system using the Agile Software Development Methodology. The development process includes planning, requirements analysis, system design, system development, testing, deployment, and maintenance. Throughout these phases, the researchers continuously evaluated and refined the system based on user feedback. During system operation, processes such as user authentication, request submission, request processing, approval or rejection of requests, records management, request tracking, notification generation, report generation, dashboard monitoring, and audit logging ensure efficient and secure administrative service delivery.

The Output is the completed i-AdminConnect web-based application that provides an integrated platform for managing administrative services and institutional records. The system enables students and teachers to submit and monitor requests online while allowing administrators to efficiently process requests, manage records, generate reports, and monitor transactions through an interactive dashboard. The implementation of the system is expected to improve administrative efficiency, reduce paperwork, enhance transparency and accountability, strengthen communication among stakeholders, and provide faster and more reliable delivery of administrative services.

Overall, the Input–Process–Output framework demonstrates how the identified resources and system requirements are transformed through systematic software development into a functional web-based administrative services and records tracking system that addresses the needs of Taguig Science High School.

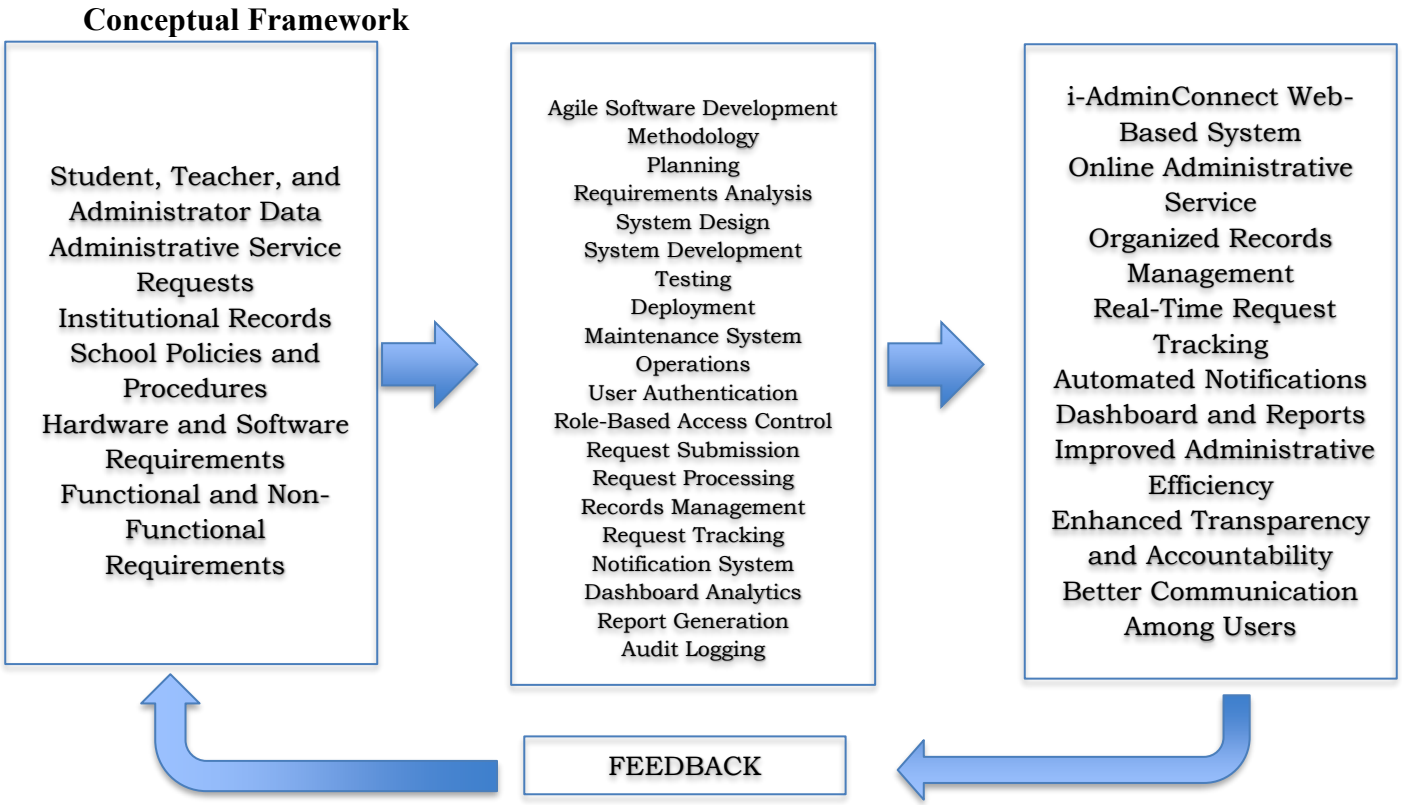



**FIGURE 1.** *Conceptual Framework*

The conceptual framework of this study illustrates the relationship among the key components that enable the development and implementation of **i-AdminConnect: A Web-Based Administrative Services and Records Tracking System for Teachers and Students**. This framework demonstrates how web-based technologies, records management, request tracking, and role-based access control are integrated to automate and improve the delivery of administrative services within an educational institution. The system aims to provide a centralized platform where students and teachers can conveniently submit administrative requests, monitor transaction status, and access authorized records, while enabling administrators to efficiently process requests, manage institutional records, and generate reports.

# CHAPTER II

## REVIEW OF RELATED LITERATURE

This chapter presents a review of local and international literature and studies that hold significant bearing on the development of i-AdminConnect: A Web-Based Administrative Services and Records Tracking System for Teachers and Students. The selected literature highlights the evolution of digital school management, the necessity of automated record tracking, and the impact of these technologies on administrative efficiency.

Educational institutions in the Philippines have increasingly adopted web-based information systems to improve the efficiency of administrative processes and record management. These studies demonstrate how automation, centralized databases, and digital platforms contribute to enhancing service delivery and reducing the limitations of manual procedures.

Contreras and Contreras (2023) developed the Automated, Systematized Enrolment Program (ASEP) to automate the enrollment process through real-time data gathering and a centralized Learner Information System (LIS). The study addressed common problems associated with manual enrollment, such as repetitive data encoding, delays in data retrieval, and inconsistencies in student records. The researchers found that automation significantly reduced manual errors, improved the accuracy of student information, and accelerated enrollment transactions. Their findings demonstrate the value of centralized information systems in improving

administrative efficiency and ensuring reliable data management within educational institutions.

Similarly, Bagacay, Talatala, and Soberano (2024) proposed an automated school registration system for registrar offices to replace traditional manual registration procedures. Their system utilized a centralized database to organize and manage student records more efficiently. The researchers emphasized that digitalizing registrar operations improved record retrieval, minimized paperwork, enhanced collaboration among administrative personnel, and increased institutional productivity. This study highlights the importance of centralized databases in supporting effective records management within educational institutions.

Taña, Teleron, and Sulima (2025) developed and evaluated a Web-Based School Management Information System with Data Analytics. The study integrated administrative functions with analytical tools that enabled school administrators to monitor institutional performance using organized and accurate information. The researchers concluded that incorporating data analytics into school management systems improved decision-making, operational monitoring, and the overall management of school information. Their findings demonstrate that modern information systems should not only automate processes but also provide meaningful insights that support administrative planning.

Oliveros, Salcedo, Tobias, and Melendez (2026) designed a Web-Based Student Portal System for Dasol Catholic School to improve accessibility and academic management. The portal enabled students to access institutional

information and services through an online platform, reducing dependence on face-to-face transactions. The study concluded that web-based portals improve communication between students and school personnel while increasing accessibility to educational services and records. These findings support the adoption of online platforms that simplify administrative interactions and improve user convenience.

Another significant local study is School-IntegrIS, an integrated school management system developed for a state college in Northern Mindanao. The study emphasized the integration of multiple administrative functions into a single information system to improve institutional efficiency and data consistency. By consolidating various school operations into one centralized platform, the researchers demonstrated how integrated systems reduce redundant processes, improve coordination among departments, and strengthen institutional data management.

Collectively, these Philippine studies demonstrate that web-based administrative systems improve efficiency, promote centralized records management, reduce manual errors, and strengthen communication among stakeholders. These findings provide a strong foundation for the development of i-AdminConnect, which seeks to integrate administrative service requests, records tracking, automated notifications, and role-based access control into a centralized web-based platform.

International research similarly emphasizes the importance of web-based school management systems in improving administrative efficiency, enhancing

accessibility, and supporting institutional decision-making through digital transformation.

Lontaan and Sinadia (2024) developed a Web-Based School Information System that streamlined school administrative operations by centralizing student and institutional information. The study highlighted the importance of digital platforms in improving information accessibility, reducing redundant administrative tasks, and facilitating efficient communication among school stakeholders. The researchers concluded that centralized web-based systems significantly improve the management of educational information.

Shaherawala, Kalariya, and Bhalodiya (2025) conducted A Comprehensive Survey on Modern School Management Systems, examining various technologies, implementation strategies, and architectural designs used in educational institutions. Their review concluded that modern school management systems should incorporate secure user authentication, centralized databases, modular design, and user-friendly interfaces to support effective administration. The study emphasized that well-designed information systems improve institutional efficiency, data security, and long-term scalability.

Dhanasekar, Geetha, Asifa, and Lakshmi (2025) developed a Web-Based School Management Application designed to improve both administrative and academic efficiency. Their research demonstrated that integrating multiple administrative functions into a single application reduced processing time, minimized manual work, and enhanced coordination among school personnel. The

researchers concluded that web-based applications significantly improve service delivery and organizational productivity within educational institutions.

Syed (2024) presented the School College Management System, which focused on improving operational consistency and institutional growth through integrated administrative processes. The study emphasized standardized workflows, centralized information management, and systematic coordination among various departments. According to the researcher, digital management systems contribute to organizational efficiency by simplifying routine administrative operations and ensuring consistency in institutional procedures.

Dhopre and Nagargoje (2026) proposed An Efficient Web-Based School Management System for Educational Institutions, which demonstrated how integrated online systems enhance institutional performance by providing centralized data management, secure access to information, and efficient administrative services. Their study highlighted the importance of scalable web technologies in supporting educational institutions as they continue to modernize administrative operations and improve user satisfaction.

Overall, international studies consistently show that web-based management systems improve administrative efficiency, strengthen data security, facilitate centralized information management, and enhance communication among users. These findings reinforce the need for integrated systems such as i-AdminConnect, which combines administrative service requests, records tracking,

role-based access control, and automated notifications into a unified web-based platform suitable for educational institutions.

Both Philippine and international studies demonstrate that web-based school management systems have become essential tools for improving administrative efficiency and institutional effectiveness. Local studies primarily focus on automating enrollment, registration, and student record management, while international studies emphasize comprehensive system integration, centralized databases, information security, and scalability. Despite these advancements, relatively few studies specifically integrate administrative service requests, real-time request tracking, automated notifications, and role-based access control into a single platform for both teachers and students. This research addresses that gap through the development of i-AdminConnect, a web-based administrative services and records tracking system designed to provide secure, transparent, and efficient administrative support within educational institutions.

**Synthesis**

The reviewed literature and related studies emphasize the importance of web-based information systems in improving administrative processes within educational institutions. Previous studies have shown that digital systems enhance efficiency by reducing processing time, improving records management, increasing data accuracy, and providing convenient access to administrative services.

The literature also highlights the value of secure features such as user authentication, role-based access control, and centralized databases in ensuring the confidentiality and integrity of institutional records. In addition, the Agile Software Development Methodology was identified as an effective approach for developing systems through continuous improvement and user feedback.

Although existing studies have focused on school management and records systems, few provide an integrated platform that combines online administrative service requests, real-time request tracking, automated notifications, and centralized records management. This research addresses that gap through the development of **i-AdminConnect**, which aims to improve the efficiency, transparency, and accessibility of administrative services at **Taguig Science High School**. The reviewed literature provides a strong foundation for the design and implementation of the proposed system.

**(Links and References will be provided after Chapter 5 in References section or Bibliography)**

# CHAPTER III

# RESEARCH METHODOLOGY

## 3.1 Research Design

This study employed the Research and Development (R&D) research design using the Agile Software Development Methodology in the design, development, implementation, and evaluation of the proposed i-AdminConnect: A Web-Based Administrative Services and Records Tracking System for Teachers and Students. The Research and Development approach is appropriate because the primary objective of the study is to develop a functional software application that addresses the administrative service needs of an educational institution while evaluating its quality and usability.

The Agile Software Development Methodology was selected because it supports iterative development, continuous testing, and regular feedback from stakeholders. Instead of completing the entire system in one phase, the researchers developed the application through several iterations or sprints, allowing improvements to be incorporated throughout the development process. This methodology ensured that the system met the requirements of administrators, teachers, and students while maintaining flexibility in responding to necessary modifications.

### 3.2 Software Development Methodology

The development of the proposed system followed the Agile Software Development Methodology consisting of six major phases.

Planning

The planning phase involved identifying the existing problems encountered in processing administrative service requests and records management. Information was gathered through interviews, observation, and document review. The researchers identified the objectives, scope, system requirements, and expected outputs of the proposed system. The modules to be included in the system were also determined during this phase.

Requirements Analysis

During this phase, the researchers analyzed both the functional and non-functional requirements of the proposed system.

**Functional Requirements**

The proposed system includes the following major functions:

- Secure user authentication
- Role-Based Access Control (RBAC)
- Student request submission
- Teacher request submission
- Administrative request processing
- Request approval and rejection
- Records management

- Real-time request tracking
- Automated notifications
- Dashboard analytics
- Report generation
- Audit logging

**Non-Functional Requirements**

The system was designed to satisfy the following quality requirements:

- Security
- Reliability
- Performance
- Scalability
- Availability
- Maintainability
- User-friendliness
- System Design

After analyzing the system requirements, the researchers prepared the design specifications of the proposed system.

The design phase included the preparation of:

- System Architecture
- Entity Relationship Diagram (ERD)
- Data Flow Diagram (DFD)
- Database Design
- Navigation Structure

- System Flowcharts

The proposed system follows a Three-Tier Client-Server Architecture, consisting of the Presentation Layer, Application Layer, and Database Layer. This architecture separates user interaction, business logic, and data management to improve system performance, maintainability, and security.

**Development**

The system was developed using the following technologies:

| Software Component | Technology Used |
|---|---|
| Front-end | HTML5, CSS3, Bootstrap, JavaScript |
| Back-end | PHP |
| Database | MySQL |
| Web Server | Apache (XAMPP) |
| Code Editor | Visual Studio Code |
| Browser | Google Chrome / Microsoft Edge |

The following modules were developed:

- Login Module
- User Management
- Student Module
- Teacher Module
- Administrator Module
- Administrative Request Module
- Records Management Module
- Notification Module

- Dashboard Analytics
- Reports Module
- Audit Trail Module
- Testing

The completed system underwent several testing procedures before implementation.

Unit Testing verified that each module performed its intended function correctly.

Integration Testing ensured that all modules communicated properly.

System Testing verified that the entire system operated according to the specified requirements.

User Acceptance Testing (UAT) was conducted among administrators, teachers, and students to determine the acceptability of the developed system.

Deployment and Maintenance

After successful testing, the system was deployed for demonstration and evaluation within the selected educational institution. Necessary corrections, updates, and improvements were performed based on the comments and recommendations of the evaluators.

### 3.3 Research Locale

The study was conducted at Taguig Science High School, where the existing administrative services for students and teachers are still largely handled through

manual procedures. The institution served as the primary source of system requirements, respondents, and evaluation of the proposed web-based administrative services and records tracking system.

### 3.4 Population of the Study

The respondents of the study consisted of students from Grades 10 to 12, teaching staff, and school administrators of Taguig Science High School. These groups were selected because they are the intended users of the proposed system.

The total population of the study was 737 respondents, distributed as follows:

**TAGUIG SCIENCE HIGH SCHOOL**

| GRADE | MALE | FEMALE | TOTAL |
|---|---|---|---|
| e | 100 | 108 | 208 |
| Grade 11 – STEM | 102 | 100 | 202 |
| Grade 11 – ABM | 7 | 28 | 35 |
| Grade 12 – STEM | 100 | 115 | 215 |
| Grade 12 – ABM | 6 | 12 | 18 |
| Teaching Staff | 21 | 29 | 50 |
| Administrator | 4 | 5 | 9 |
| **GRAND TOTAL** | **340** | **397** | **737** |

*Source: Registrar's Office, Taguig Science High School (2026)*

### 3.5 Sampling Technique and Sample Size

The study employed proportionate stratified random sampling to ensure that each respondent group was fairly represented. The respondents were divided into strata according to grade level, teaching staff, and administrators.

The sample size was determined using Slovin's Formula with a 5% margin of error.

Substituting the values:

n=

1+737(0.05)

2

737

n=

1+737(0.0025)

737

n=

2.8425

737

n=259.27

The computed sample size was rounded to 260 respondents.

Using proportional allocation, the distribution of respondents is shown below.

| Respondent Group | Population | Sample Size |
|---|---|---|
| Grade 10 | 208 | 73 |
| Grade 11 – STEM | 202 | 71 |
| Grade 11 – ABM | 35 | 12 |
| Grade 12 – STEM | 215 | 76 |
| Grade 12 – ABM | 18 | 6 |
| Teaching Staff | 50 | 18 |
| Administrators | 9 | 4 |
| **Total** | **737** | **260** |

### 3.6 Data Gathering Procedure

The researchers followed the procedures below in gathering the necessary data.

Interview

Interviews were conducted with school administrators to identify existing administrative procedures, document processing workflows, and current challenges.

Observation

The researchers observed how administrative services were performed, including the submission of requests, records management, and document releasing procedures.

Document Review

Existing request forms, institutional policies, and administrative records were examined to identify system requirements.

System Development

The proposed system was developed following the Agile Software Development Methodology.

### System Evaluation

Upon completion, the respondents evaluated the system using a standardized evaluation questionnaire based on ISO/IEC 25010 Software Quality Standards.

### 3.7 Research Instrument

The primary instrument used in evaluating the developed system was a structured questionnaire based on the ISO/IEC 25010 Software Product Quality Model.

The questionnaire measured the following software quality characteristics:

- Functional Suitability
- Performance Efficiency
- Compatibility
- Usability
- Reliability
- Security
- Maintainability
- Portability

A four-point Likert Scale was used.

| Scale | Interpretation |
|---|---|
| 5 | Strongly Agree |
| 3 | Agree |
| 2 | Disagree |
| 1 | Strongly Disagree |

3.8 Statistical Treatment of Data

The following statistical tools were used in the study.

- Frequency and Percentage
- Frequency and percentage were used to describe the demographic profile of the respondents.
- Percentage=Total
- Frequency×100
- Weighted Mean

The weighted mean was used to determine the level of acceptability of the developed system.

$$WM = \frac{\sum fx}{N}$$

where:

WM = Weighted Mean

f = Frequency

x = Rating

N = Total Number of Responses

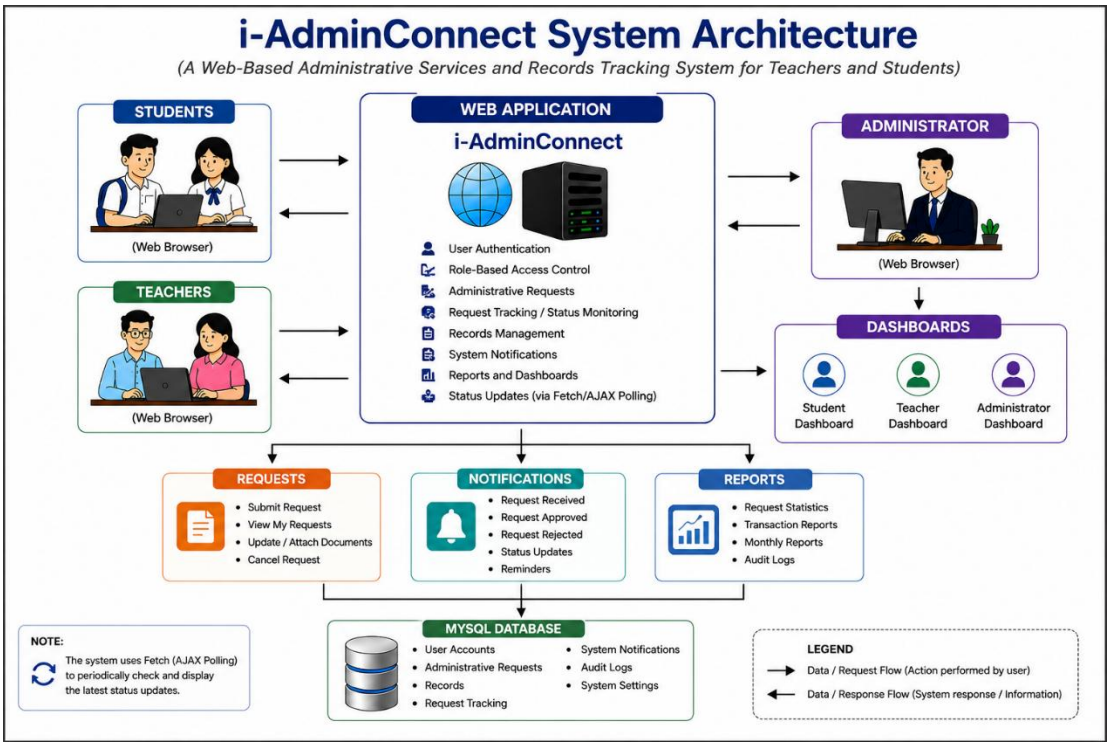
The weighted mean was interpreted using the following scale:

| Mean Range | Interpretation |
|---|---|
| 4.21–5.00 | Excellent |
| 3.41–4.20 | Very Good |
| 2.61–3.40 | Good |
| 1.81–2.60 | Fair |
| 1.00–1.80 | Poor |

### 3.9 Ethical Considerations

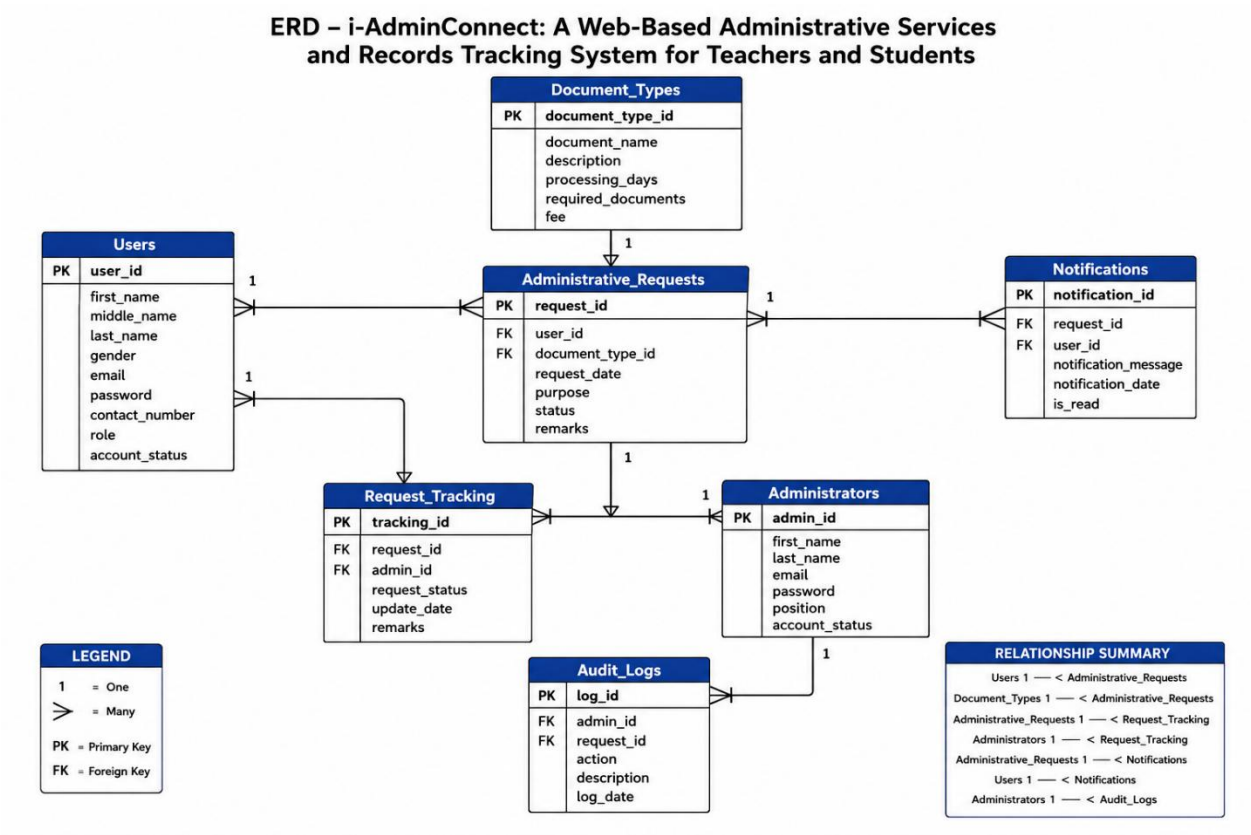
The researchers observed ethical principles throughout the conduct of the study. Permission to conduct the research was obtained from the school administration before gathering data. Participation in the evaluation was voluntary, and respondents were informed of the purpose of the study. The identities and responses of participants were kept confidential and were used solely for academic purposes. All collected data were securely stored and handled in accordance with applicable data privacy policies.

**System Architecture**



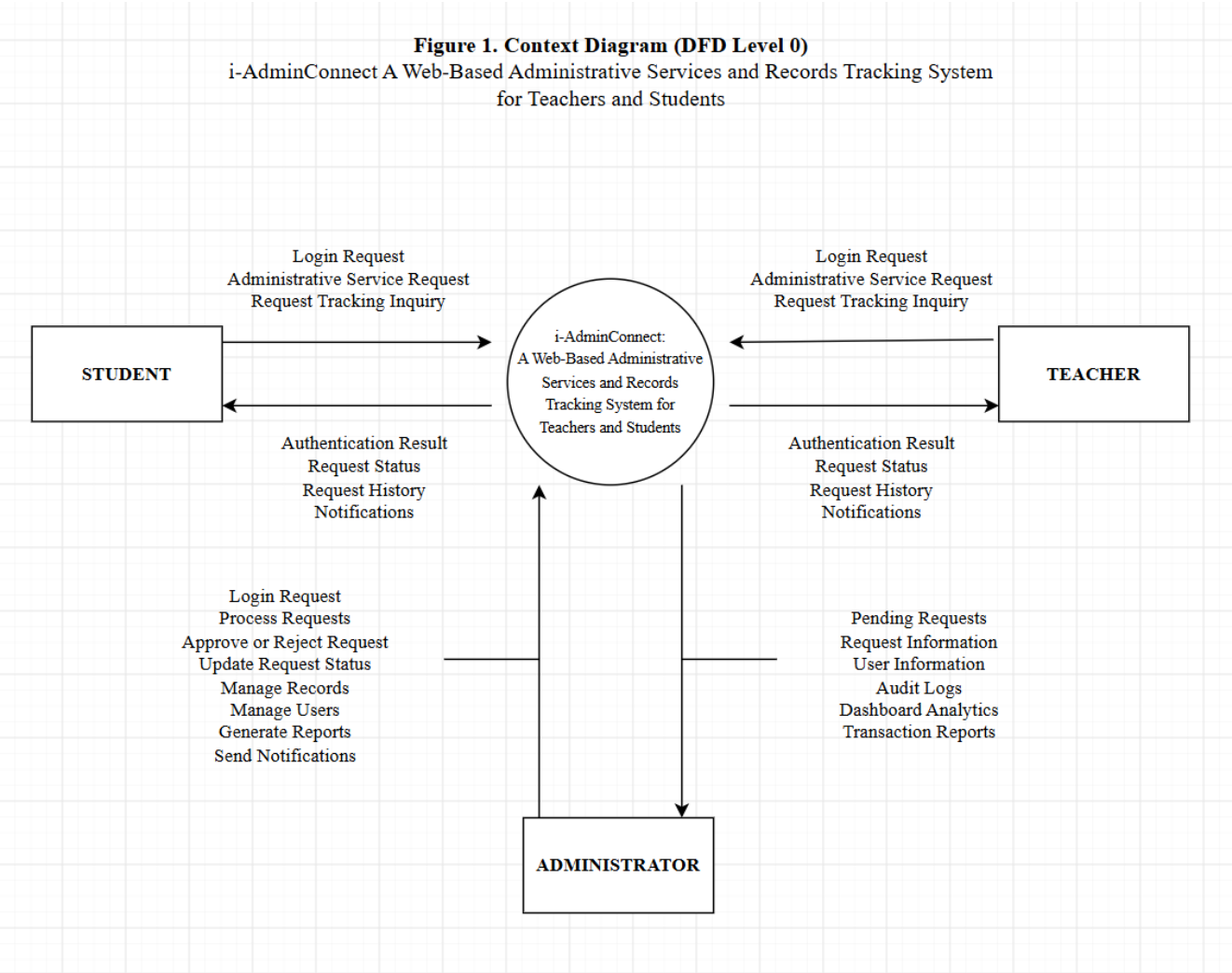

This figure illustrates the overall architecture of the proposed i-AdminConnect system, showing the interactions among students, teachers, administrators, the web application, the database, the request management module, the notification module, and the reporting module. The system uses role-based access control and request tracking to facilitate the efficient processing of administrative requests.

## Entity Relationship Diagram (ERD)




The Entity Relationship Diagram (ERD) illustrates the database structure of the proposed **i-AdminConnect: A Web-Based Administrative Services and Records Tracking System for Teachers and Students**. It presents the system's entities, attributes, primary keys, foreign keys, and the relationships among the database tables, showing how user accounts, document types, administrative requests, request tracking, notifications, administrator records, and audit logs are organized to support the system's core functions, maintain data integrity, and ensure efficient data storage and retrieval.

**Figure 1. Context Diagram (DFD Level 0)**
i-AdminConnect A Web-Based Administrative Services and Records Tracking System for Teachers and Students

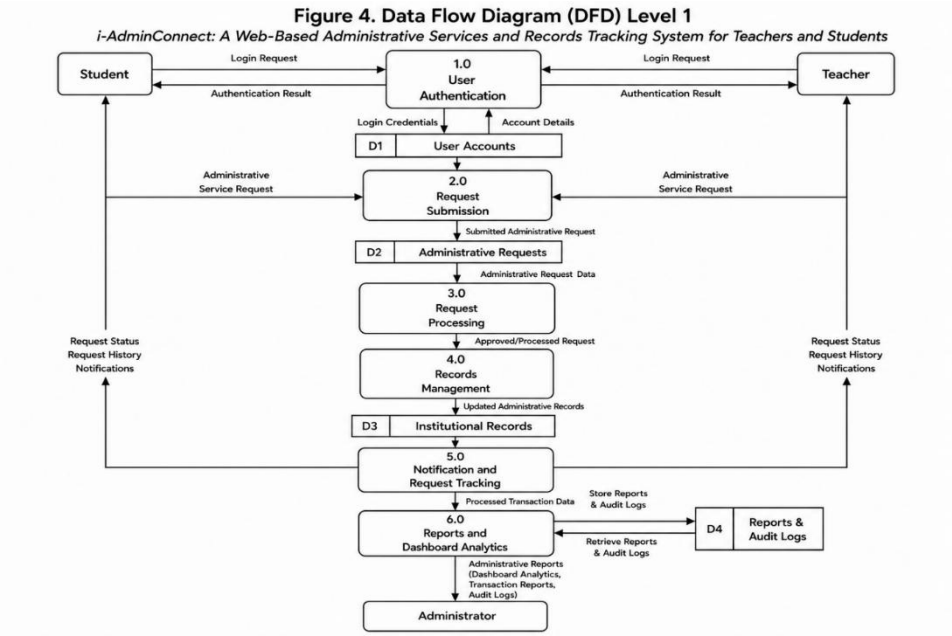



The **Level 0 Data Flow Diagram (DFD)** presents a high-level overview of the proposed **i-AdminConnect: A Web-Based Administrative Services and Records Tracking System for Teachers and Students**. The system interacts with three external entities: **Student**, **Teacher**, and **Administrator**. Students and teachers log in to the system, submit administrative service requests, inquire about the status of their requests, and receive authentication results, request status updates, request history, and automated notifications. The **Administrator** manages the overall operation of the system by processing requests, approving or rejecting applications, updating request statuses,

managing records and user accounts, generating reports, and sending notifications. In return, the system provides the administrator with pending requests, request information, user information, dashboard analytics, audit logs, and transaction reports to support efficient monitoring and decision-making. Overall, the system serves as a centralized platform that streamlines administrative services, improves request tracking, enhances communication among users, and promotes efficient and transparent records management.

**Figure 4. Data Flow Diagram (DFD) Level 1**

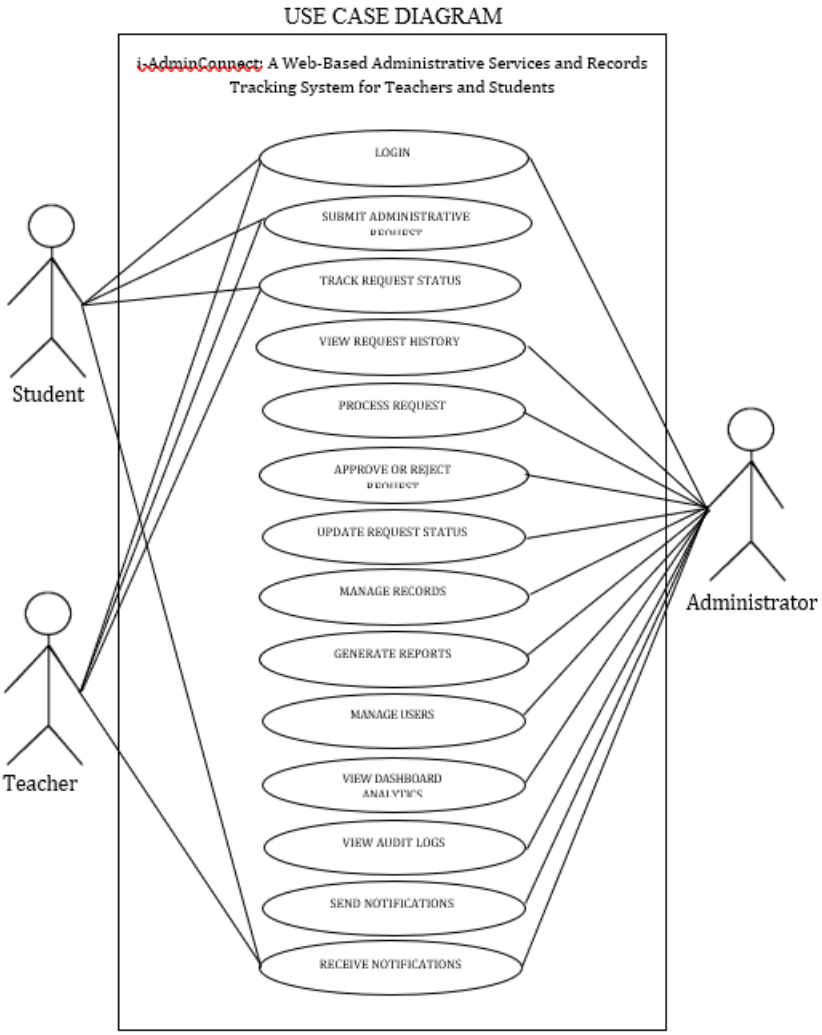
*i-AdminConnect: A Web-Based Administrative Services and Records Tracking System for Teachers and Students*




The Data Flow Diagram (DFD) Level 1 illustrates the detailed flow of information within the proposed i-AdminConnect: A Web-Based Administrative Services and Records Tracking System for Teachers and Students. The process begins with Students and Teachers logging into the system through Process 1.0, User Authentication, where their credentials are validated using Data Store D1 (User Accounts). After successful authentication, users submit administrative service requests through Process 2.0, Request Submission, and the requests are stored in Data Store D2 (Administrative Requests).

The requests are then reviewed and processed by the Administrator in Process 3.0, Request Processing before being updated and organized through Process 4.0, Records Management, where they are stored in Data Store D3 (Institutional Records). Process 5.0, Notification and Request Tracking, provides users with request status updates, request history, and notifications, while Process 6.0, Reports and Dashboard Analytics generates reports and audit logs stored in Data Store D4 (Reports & Audit Logs). The diagram shows how data moves through the system to support efficient request processing, records management, request tracking, and administrative reporting.

## User Interface Design




The Use Case Diagram presents the different interactions between the users and the proposed i-AdminConnect: A Web-Based Administrative Services and Records Tracking System for Teachers and Students. It identifies the major functions of the system and shows how each user can access the features based on their assigned role.

The system has three primary users: the Student**,** Teacher, and Administrator**.**

Students can log in to the system using their authorized accounts to access the available services. After logging in, they can submit requests for administrative documents, check the current status

of their requests, view their previous transactions through the request history, and receive notifications whenever there are updates regarding their requests.

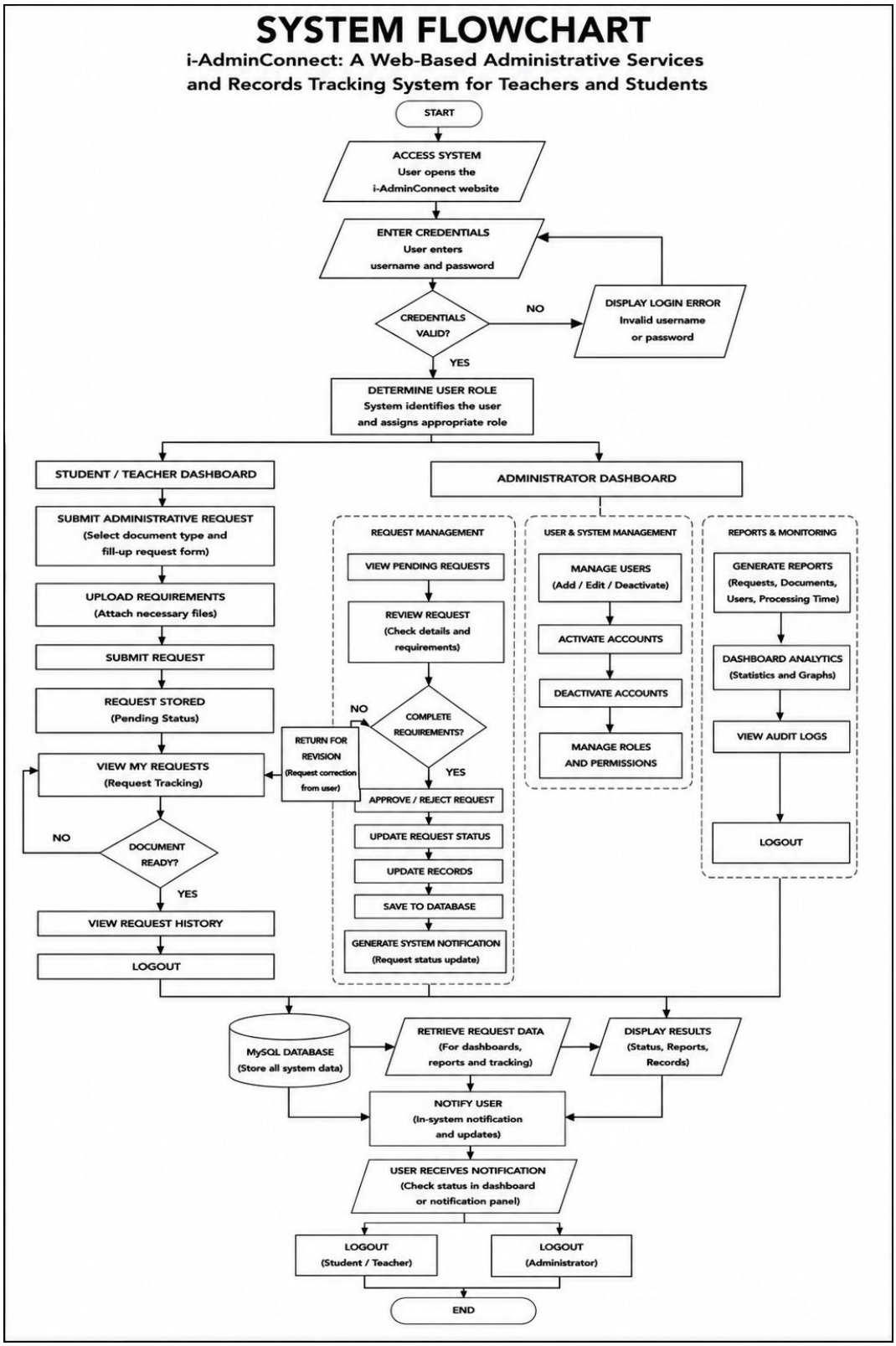
Teachers also have access to the same core functions. They can submit requests for official school documents such as Good Moral Certificates, Certification of Employment (COE), Certification of Ratings, Service Records, Leave Credit Certifications, and Service Credit Certifications. In addition, teachers can monitor the progress of their requests, review their request history, and receive system notifications whenever the administrative office updates the status of their transactions.

The Administrator is responsible for managing the overall operation of the system. The administrator receives and processes submitted requests, approves or rejects applications, updates the status of requests, manages institutional records, and maintains user accounts. The administrator can also generate reports, view dashboard analytics, monitor audit logs, and send notifications to inform teachers and students about the progress or completion of their requests.

The Use Case Diagram shows how the proposed system supports a centralized and organized process for handling administrative services. By allowing teachers and students to submit and monitor their requests online while enabling administrators to manage transactions efficiently, the system helps reduce paperwork, minimize lost or misplaced requests, lessen unnecessary follow-up visits to the administrative office, and improve the overall quality and efficiency of administrative services within the institution.

# SYSTEM FLOWCHART

## i-AdminConnect: A Web-Based Administrative Services and Records Tracking System for Teachers and Students



The System Flowchart illustrates the step-by-step operational process of the proposed **i-AdminConnect: A Web-Based Administrative Services and Records Tracking System for Teachers and Students**. The process begins when a user accesses the system and logs in using valid credentials. After successful authentication, the system determines the user's role and directs them to the appropriate dashboard.

Students and teachers can submit administrative requests, upload the required documents, monitor the status of their requests, receive system notifications, and view their request history. On the other hand, the administrator manages user accounts, reviews and validates submitted requests, approves or rejects requests, updates request statuses, generates reports, and maintains audit logs. All transactions are processed and stored in the MySQL database, enabling accurate request tracking, records management, and report generation. The flowchart demonstrates the logical sequence of system processes, user interactions, decision points, and data flow from the beginning of the transaction until the completion of the request.

**Data Analysis**

**Quantitative Data Analysis**

The quantitative data were obtained from 314 completed survey questionnaires collected through Google Forms from students and other stakeholders.

The survey contained statements regarding:

- Current Administrative Process
- Functionality
- Usability
- Efficiency
- Reliability
- Security
- Administrative Effectiveness
- Overall Acceptability

Responses were measured using a 4-point Likert Scale.

The collected responses were analyzed using:

- Frequency
- Percentage
- Weighted Mean

The weighted mean was computed to determine the respondents' level of agreement for every criterion. The resulting values were interpreted using the verbal interpretations specified in the research instrument.

The survey responses indicate that the majority of respondents agreed that the current administrative process involves delays and that there is a need to improve existing services through a web-based system. Most respondents also expressed support for features such as online request submission, request tracking, automated notifications, centralized records management, dashboard reporting, and secure storage of administrative records. Overall, the responses demonstrate strong acceptance of the proposed i-AdminConnect system as a solution to improve administrative efficiency.

**Qualitative Data Analysis**

Qualitative data were gathered through semi-structured interviews with seven (7) school personnel from the Registrar's Office, Administrative Office, Accounting Unit, and other administrative departments.

The interview responses were analyzed using thematic analysis, wherein similar responses were grouped into recurring themes.

The analysis revealed the following major themes.

**Theme 1: Manual Processing Causes Delays**

Most interviewees explained that document requests are still processed manually, resulting in slower transactions.

Representative responses include:

- document processing is somewhat slow;
- requests are manually processed;
- delays occur before documents are released.

These responses indicate that the current process requires improvement through automation.

**Theme 2: Difficulty in Tracking Requests**

Several respondents mentioned that users often return multiple times to check whether their requested documents are already available.

Common responses included:

- delayed claim dates;
- several days pass before notification;
- users repeatedly visit the office to inquire about request status.

These findings highlight the need for a real-time request tracking feature.

**Theme 3: Challenges in Records Retrieval**

Interview participants stated that retrieving previously processed documents can be time-consuming, especially when records are stored in filing cabinets or scattered across digital folders.

Frequently mentioned concerns include:

- difficulty locating old files;
- searching through physical records;
- retrieving documents from multiple storage locations.

These findings support the development of a centralized records management system.

**Theme 4: Need for Automation**

Nearly all interviewees believed that a web-based administrative services system would improve office operations.

Commonly cited benefits include:

- faster document processing;
- reduced paperwork;
- centralized storage of records;
- improved workflow;
- increased efficiency;
- better monitoring of requests.

**Theme 5: Desired System Features**

Interview participants consistently recommended several essential features for the proposed system.

Frequently mentioned features include:

- Online Request Submission
- Request Tracking
- Automated Notifications
- Dashboard Reports
- Audit Logs
- User Role Management
- Centralized Database
- Real-Time Monitoring
- Processing Time Monitoring

These recommendations were incorporated into the functional requirements of the proposed system.

**Summary of Findings**

Based on the survey and interview responses, several important findings emerged.

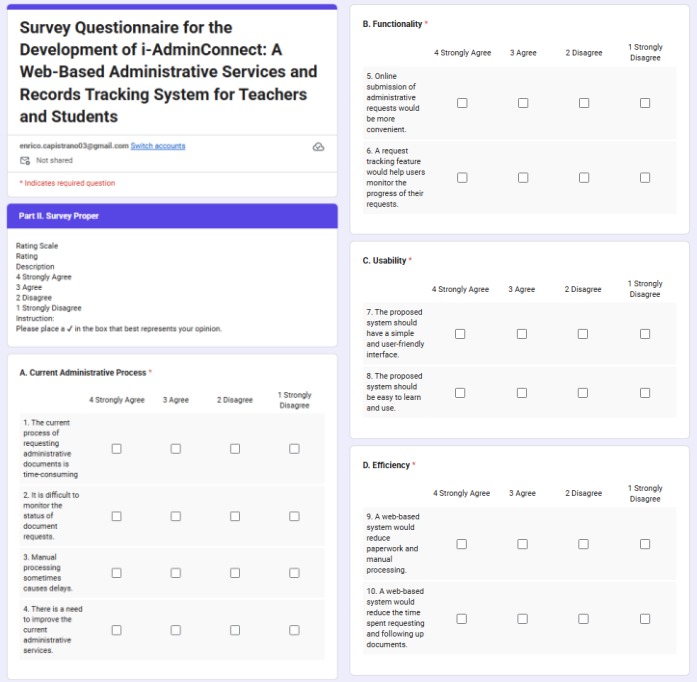
First, respondents acknowledged that the existing administrative process relies heavily on manual procedures, resulting in delays in processing and releasing requested documents.

Second, respondents identified difficulties in monitoring the status of document requests and retrieving previously processed records. These issues often require repeated follow-up visits and additional processing time.

Third, both survey respondents and interview participants expressed strong support for the implementation of a web-based administrative services and records tracking system. Features such as online request submission, request tracking, automated notifications, centralized records management, and dashboard reporting were consistently identified as important components of the proposed system.

Finally, the findings demonstrate that the proposed i-AdminConnect system addresses the operational needs of Taguig Science High School by improving efficiency, enhancing transparency, reducing paperwork, strengthening records management, and providing faster administrative services to students, teachers, and administrators.

## Questionnaires



Survey Questionnaire for the Development of i-AdminConnect: A Web-Based Administrative Services and Records Tracking System for Teachers and Students

enrico.capistrano03@gmail.com Switch accounts

Not shared

* Indicates required question

Part II. Survey Proper

Rating Scale
Rating
Description
4 Strongly Agree
3 Agree
2 Disagree
1 Strongly Disagree
Instruction:
Please place a ✓ in the box that best represents your opinion.

A. Current Administrative Process *

| | 4 Strongly Agree | 3 Agree | 2 Disagree | 1 Strongly Disagree |
|---|---|---|---|---|
| 1. The current process of requesting administrative documents is time-consuming | | | | |
| 2. It is difficult to monitor the status of document requests. | | | | |
| 3. Manual processing sometimes causes delays. | | | | |
| 4. There is a need to improve the current administrative services. | | | | |

B. Functionality *

| | 4 Strongly Agree | 3 Agree | 2 Disagree | 1 Strongly Disagree |
|---|---|---|---|---|
| 5. Online submission of administrative requests would be more convenient. | | | | |
| 6. A request tracking feature would help users monitor the progress of their requests. | | | | |

C. Usability *

| | 4 Strongly Agree | 3 Agree | 2 Disagree | 1 Strongly Disagree |
|---|---|---|---|---|
| 7. The proposed system should have a simple and user-friendly interface. | | | | |
| 8. The proposed system should be easy to learn and use. | | | | |

D. Efficiency *

| | 4 Strongly Agree | 3 Agree | 2 Disagree | 1 Strongly Disagree |
|---|---|---|---|---|
| 9. A web-based system would reduce paperwork and manual processing. | | | | |
| 10. A web-based system would reduce the time spent requesting and following up documents. | | | | |

E. Reliability *

| | 4 Strongly Agree | 3 Agree | 2 Disagree | 1 Strongly Disagree |
|---|---|---|---|---|
| 11. A centralized records management system would help prevent lost or misplaced records. | | | | |

F. Security *

| | 4 Strongly Agree | 3 Agree | 2 Disagree | 1 Strongly Disagree |
|---|---|---|---|---|
| 12. It is important that administrative records and personal information are securely protected. | | | | |

G. Administrative Effectiveness *

| | 4 Strongly Agree | 3 Agree | 2 Disagree | 1 Strongly Disagree |
|---|---|---|---|---|
| 13. Automated notifications would reduce unnecessary follow-up visits to the administrative office. | | | | |
| 14. Dashboard reports and centralized records would help administrators manage requests more efficiently. | | | | |

H. Overall Acceptability *

| | 4 Strongly Agree | 3 Agree | 2 Disagree | 1 Strongly Disagree |
|---|---|---|---|---|
| 15. I support the development and implementation of i-AdminConnect in our institution. | | | | |

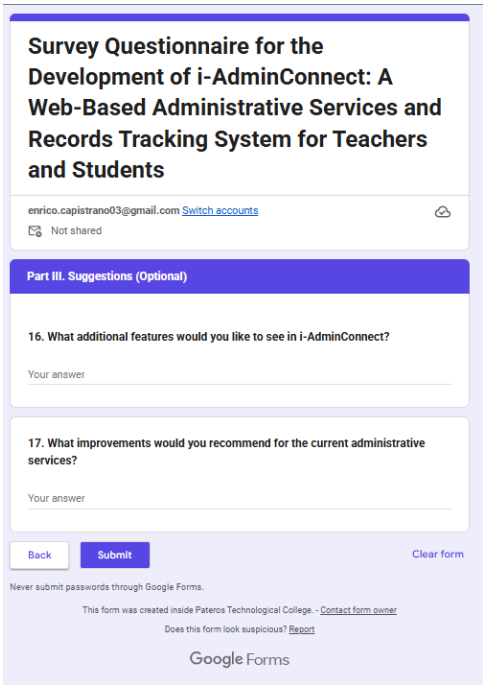
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Survey Questionnaire for the Development of i-AdminConnect: A Web-Based Administrative Services and Records Tracking System for Teachers and Students

enrico.capistrano03@gmail.com Switch accounts

Not shared

Part III. Suggestions (Optional)

16. What additional features would you like to see in i-AdminConnect?

Your answer

17. What improvements would you recommend for the current administrative services?

Your answer

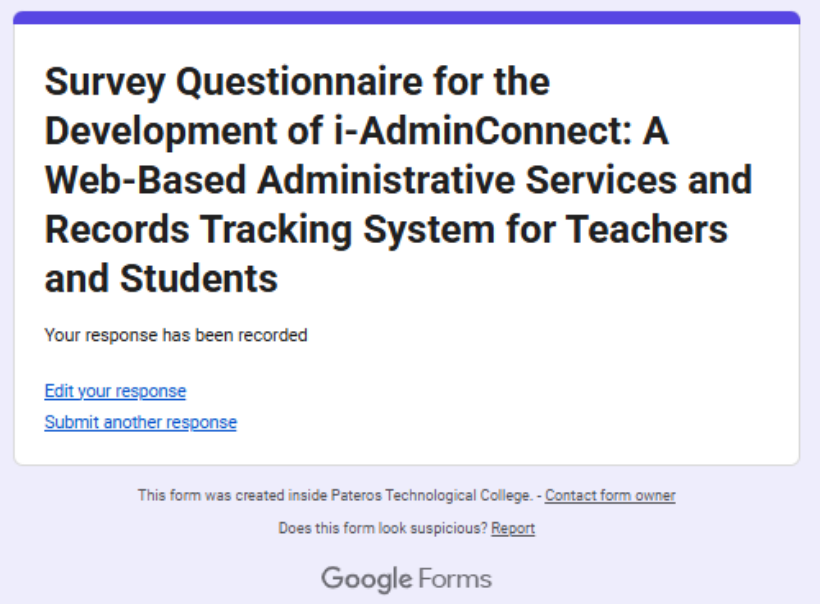
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Survey Questionnaire for the Development of i-AdminConnect: A Web-Based Administrative Services and Records Tracking System for Teachers and Students

Your response has been recorded

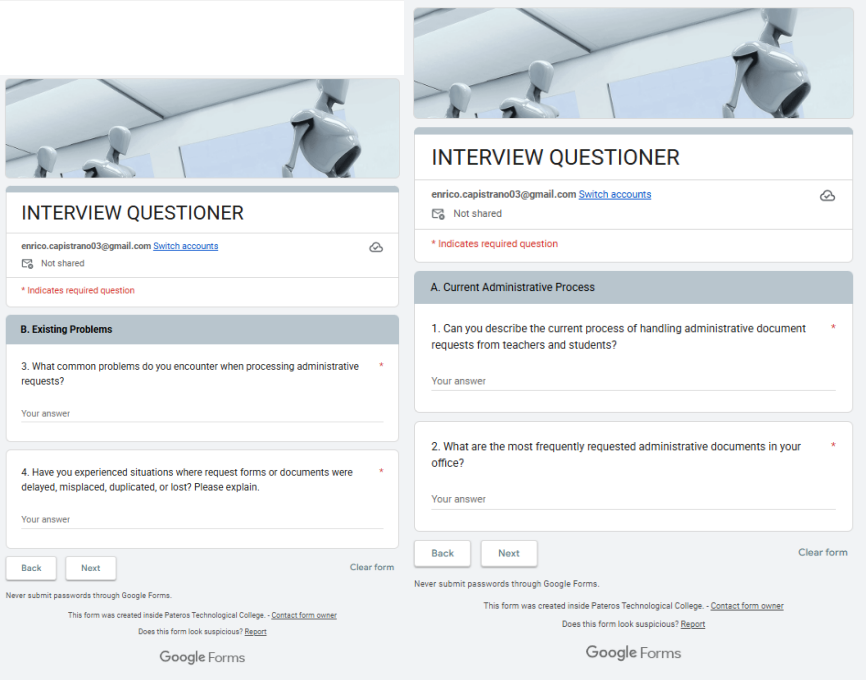
Edit your response

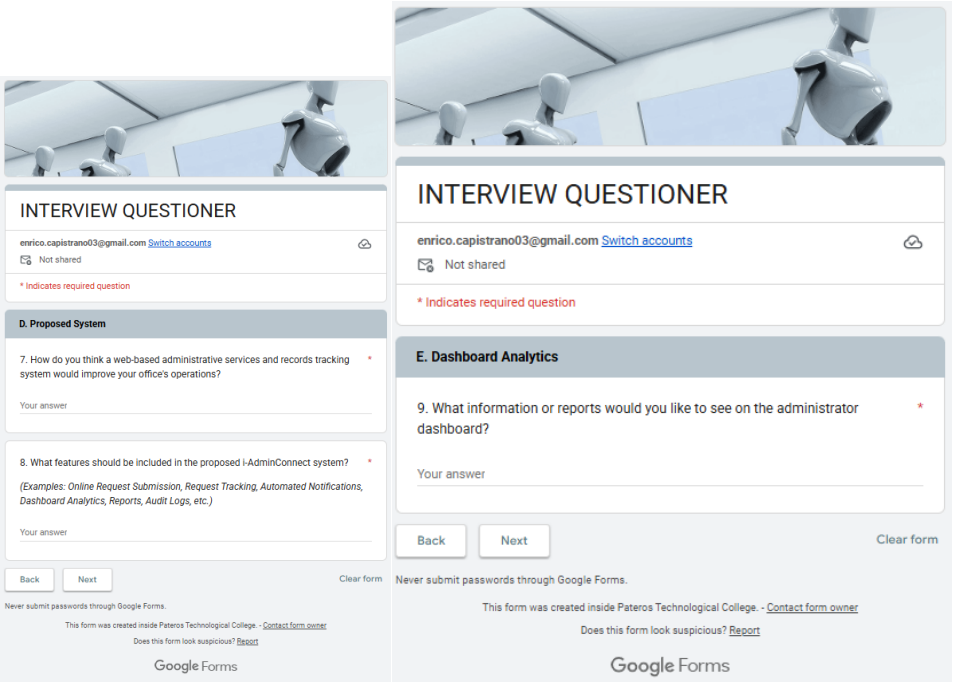
Submit another response

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INTERVIEW QUESTIONER
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* Indicates required question
B. Existing Problems
3. What common problems do you encounter when processing administrative requests? *
Your answer
4. Have you experienced situations where request forms or documents were delayed, misplaced, duplicated, or lost? Please explain. *
Your answer
Back
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INTERVIEW QUESTIONER
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* Indicates required question
A. Current Administrative Process
1. Can you describe the current process of handling administrative document requests from teachers and students? *
Your answer
2. What are the most frequently requested administrative documents in your office? *
Your answer
Back
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INTERVIEW QUESTIONER
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Not shared
* Indicates required question
D. Proposed System
7. How do you think a web-based administrative services and records tracking system would improve your office's operations? *
Your answer
8. What features should be included in the proposed I-AdminConnect system? *
(Examples: Online Request Submission, Request Tracking, Automated Notifications, Dashboard Analytics, Reports, Audit Logs, etc.)
Your answer
Back
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INTERVIEW QUESTIONER
enrico.capistrano03@gmail.com Switch accounts
Not shared
* Indicates required question
E. Dashboard Analytics
9. What information or reports would you like to see on the administrator dashboard? *
Your answer
Back
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## Documentation of Data Gathering Activities

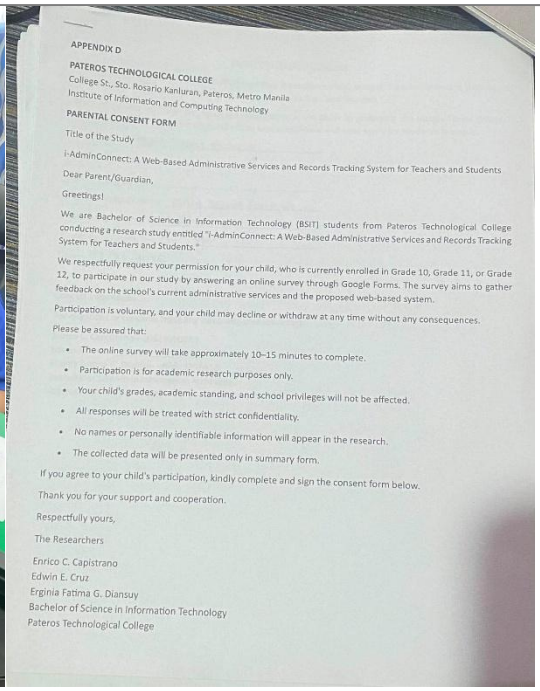










APPENDIX D

**PATEROS TECHNOLOGICAL COLLEGE**
College St., Sto. Rosario Kanluran, Pateros, Metro Manila
Institute of Information and Computing Technology

**PARENTAL CONSENT FORM**

Title of the Study

i-AdminConnect: A Web-Based Administrative Services and Records Tracking System for Teachers and Students

Dear Parent/Guardian,

Greetings!

We are Bachelor of Science in Information Technology (BSIT) students from Pateros Technological College conducting a research study entitled "i-AdminConnect: A Web-Based Administrative Services and Records Tracking System for Teachers and Students."

We respectfully request your permission for your child, who is currently enrolled in Grade 10, Grade 11, or Grade 12, to participate in our study by answering an online survey through Google Forms. The survey aims to gather feedback on the school's current administrative services and the proposed web-based system.

Participation is voluntary, and your child may decline or withdraw at any time without any consequences.

Please be assured that:

- The online survey will take approximately 10–15 minutes to complete.
- Participation is for academic research purposes only.
- Your child's grades, academic standing, and school privileges will not be affected.
- All responses will be treated with strict confidentiality.
- No names or personally identifiable information will appear in the research.
- The collected data will be presented only in summary form.

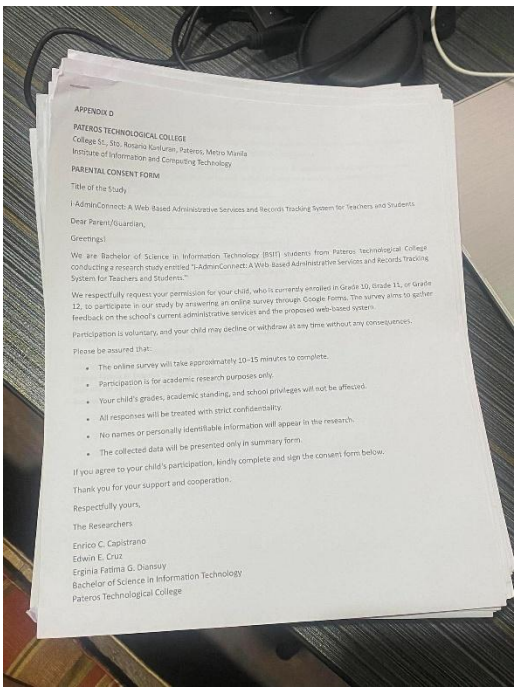
If you agree to your child's participation, kindly complete and sign the consent form below.

Thank you for your support and cooperation.

Respectfully yours,

The Researchers

Enrico C. Capistrano
Edwin E. Cruz
Erginia Fatima G. Diansuy
Bachelor of Science in Information Technology
Pateros Technological College



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